

Multiple Choice (10 marks)

1. In a healthy person maintaining energy balance in equilibrium (i.e. mathematical equality of energy intake and energy expenditure) is generally achieved on a:
 - (a) 4-5 day basis
 - (b) 3-4 day basis
 - (c) 5-6 day basis
 - (d) Hourly basis
 - (e) Monthly basis
2. If links between various diseases are found, then future older adults will probably
 - (a) Have a shorter life expectancy
 - (b) Be less likely to seek medical advice
 - (c) Take fewer medications
 - (d) Pay far less for pharmaceuticals
 - (e) Have higher medical insurance premiums
3. Which of the following is a function of folate?
 - (a) Synthesis of fatty acids from glucose
 - (b) Conversion of oxaloacetate to phosphoenolpyruvate
 - (c) Carboxylation of pyruvate to oxaloacetate
 - (d) Conversion of fructose to glucose
 - (e) Synthesis of TMP (thymidine monophosphate)
4. A quasi species virus such as influenza and HIV has which of the following characteristics?
 - (a) Possesses only RNA
 - (b) A stable and unchanging genome
 - (c) Possesses only DNA
 - (d) Has a single, unsegmented genome
 - (e) Possesses RNA and DNA
5. The group that permits analysis of the entire natural history of HIV/AIDS is:
 - (a) Non-seroconverters
 - (b) Seroconverters

- (c) Short-term survivors with low levels of CD 4+ cells
- (d) Long-term survivors
- (e) Long-term survivors with high levels of CD 4+ cells

True / False (10 marks)

Answer True or False

6. True or False: Urea is the primary nitrogenous waste product excreted in urine from the metabolism of proteins.
7. True or False: Mouthwash containing alcohol is recommended for effective cleaning and disinfection of the mouth.
8. True or False: Mutations that cause achondroplasia can be classified as recessive negative due to their impact on bone growth regulation.
9. True or False: The mean rate of energy expenditure for a male athlete with an oxygen consumption of 2 l/min is approximately 40 kJ/min.
10. True or False: Nearly one-third of grieving individuals require some form of professional help to cope with their loss.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of papillary muscles in preventing the eversion of the right atrioventricular valve and analyze their significance in maintaining proper cardiac function.
12. Discuss the rationale behind the World Health Organization's recommendation for zinc supplementation, particularly for men with high physical activity levels, and analyze its implications for health.

End of Paper